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Stochastic Analysis of Symmetric Two-Flow Wireless-Fair Scheduling

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ABSTRACT

Future Wireless Networks are expected to offer diverse services with different grades of Quality of Service (QoS).Wireless scheduling plays an important role in QoS provisioning since it determines how scarce resources will be allocated to concurrently support the diverse demands of users.

In this report, we shall develop an analytical model for a simple symmetric two-flow wireless scheduling algorithm based on the fair queuing paradigm *(Wireless-Fair Scheduling*). By proper selection of the analysis interval, time instances as well as definition of a suitable state variable, the behavior of the scheduler can be characterized by a one dimensional Markov Chain. Based on the model, we shall derive the delay and fairness performance of the two-flow scheduler in terms of channel state parameters.

In addition, we shall also define a *Wired-Fair* scheduling scheme which is fair and but not channel-efficient and a *Channel-Efficient* scheduling scheme which is not fair. We shall compare the performance of these schemes under different channel conditions in terms of channel-efficiency and fairness.